Exercises on Generating Functions

Revised and expanded · 25 problems

Work with formal power series. For a sequence $(a_n)_{n\geq 0}$, the ordinary and exponential generating functions are, respectively,

$$\text{OGF}: \quad \sum_{n\geq 0} a_n x^n, \qquad \text{EGF}: \quad \sum_{n\geq 0} a_n \frac{x^n}{n!}.$$

A *composition* has positive, ordered parts; an *integer partition* has positive, unordered parts. Unless stated otherwise, count the empty object once. Use generating functions unless another method is requested.

1. **Bills in a wallet.** You have three 20-dollar bills, five 10-dollar bills, and six 5-dollar bills. Find a polynomial $P(x)$ such that the coefficient of x^n is the number of ways to select bills totalling n dollars, under each convention:

   (a) Bills of the same denomination are indistinguishable.

   (b) All individual bills are distinguishable, for example by their serial numbers.

   In both cases, the order in which the bills are selected is irrelevant.

2. **Compositions with odd parts.** Let c_n count compositions of n into odd parts, with $c_0 = 1$. Find its OGF and express c_n, for $n \geq 1$, in terms of the Fibonacci numbers $F_0 = 0$, $F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$. Treat $n = 0$ separately.

3. **Divisibility in a weak composition.** For $n \geq 1$, let q_n count tuples of nonnegative integers $(r_1, \ldots, r_n)$ satisfying

   $$r_1 + \cdots + r_n = n, \qquad i \mid r_i \quad (1 \leq i \leq n).$$

   Set $q_0 = 1$. Find $\sum_{n\geq 0} q_n x^n$, and deduce its relationship with the partition numbers $p(n)$. Also give a direct bijection.

4. **One red cycle and one blue cycle.** Let q_n count permutations of $\{1, \ldots, n\}$ with exactly two cycles, one coloured red and the other blue. Both cycles are nonempty, and exchanging their colours gives a different object. Cycles are considered up to cyclic rotation, as usual. Find the EGF; here $q_0 = q_1 = 0$.

5. **Partitions into odd parts.** Let $p_{\text{odd}}(n)$ count partitions of n into odd parts, with $p_{\text{odd}}(0) = 1$. Find its OGF.

6. **Partitions into distinct parts.** Let $p_{\text{dist}}(n)$ count partitions of n into pairwise distinct parts, with $p_{\text{dist}}(0) = 1$. Find its OGF.

7. **Odd parts versus distinct parts.** Compare the products in Problems 5 and 6 to prove that $p_{\text{odd}}(n) = p_{\text{dist}}(n)$ for every $n \geq 0$. Can you also give a bijective proof?

8. **Three statistics of a partition.** Let $p(n,k,\ell)$ count partitions of n with exactly ℓ parts and largest part exactly k, where $n,k,\ell \geq 1$. Find an expression for

$$P(x,y,z) = \sum_{n,k,\ell \geq 1} p(n,k,\ell) x^n y^k z^\ell.$$

   An expression as a sum over k of finite products is sufficient.

9. **Squares and partial sums.** Find a closed form for $G(x) = \sum_{n\geq 0} n^2 x^n$. Use it to find the OGF of $b_n = \sum_{i=1}^{n} i^2$, with $b_0 = 0$, and hence derive a formula for b_n.

10. **An even number of green balls.** An unlimited supply of red, blue, and green balls is available. Balls of the same colour are indistinguishable, and the order of selection is irrelevant. Let a_n count selections of n balls in which the number of green balls is even. Find a closed form for the OGF of $(a_n)_{n\geq 0}$.

11. **A weighted binomial sum.** Evaluate $\sum_{i=0}^{n} i\binom{n}{i}$ for $n \geq 1$, using a generating function or another method. Also give a counting interpretation of the sum.

12. **Central binomial coefficients.** Find a closed form for $\sum_{n\geq 0} \binom{2n}{n} x^n$.

13. **A factorial recurrence.** Let $a_0 = 2$ and $a_n - a_{n-1} = n!$ for $n \geq 1$. Find an explicit finite-sum formula for a_n. If $A(x) = \sum_{n\geq 0} a_n x^n/n!$, find a first-order differential equation and an initial condition satisfied by $A(x)$.

14. **Three committees and a president.** Assign $n \geq 1$ distinct senators to three named committees A, B, C, each senator belonging to exactly one committee. Committee A must have even size, possibly zero; B must have odd size; and C must be nonempty and have a distinguished member as president. Choose a suitable generating function and show that the number of assignments, including the choice of president, is

$$\frac{n}{4}\bigl(3^{\,n-1} - (-1)^{\,n-1}\bigr).$$

15. **People at circular tables.** Partition n distinct people into an unordered collection of nonempty groups, and seat each group around a circular table. Tables are unlabelled; rotations of a seating are identified, but reflections are not. Groups of size one or two are allowed. Use an EGF to show that there are $n!$ arrangements, including one empty arrangement for $n = 0$.

16. **Two kinds of soup.** In Problem 15, each table is also served either mushroom soup or tomato soup, with one choice for the whole table. Show that the number of arrangements is $(n+1)!$ for every $n \geq 0$.

Nine more problems

17. Binary strings with separated ones. Let a_n count binary strings of length n with no two consecutive ones, and set $a_0 = 1$. Find its OGF and a formula in terms of Fibonacci numbers. More precisely, for $0 \le k \le n$, how many such strings have exactly k ones?

18. Palindromic compositions. A composition $(a_1, \dots, a_r)$ is palindromic if $a_i = a_{r+1-i}$ for every i. Let c_n count palindromic compositions of n, with the empty composition counted for $n = 0$. Find its OGF and a simple formula for c_n. *Hint:* build the composition from its outer pairs and its possible middle part.

19. Numeral systems as infinite products. For an integer $b \ge 2$, prove the formal power series identity

$$\prod_{j \ge 0} \left(1 + x^{b^j} + x^{2b^j} + \cdots + x^{(b-1)b^j}\right) = \frac{1}{1-x}.$$

Give both an algebraic proof and an interpretation in terms of base-b representations. Explain why each coefficient of the product is well defined.

20. Removing a staircase. Fix $k \ge 1$, and let $d_k(n)$ count partitions of n into exactly k distinct positive parts. Find $\sum_{n \ge 0} d_k(n) x^n$. *Hint:* write the parts in increasing order and subtract $1, 2, \dots, k$. What kind of partition remains after zero parts are omitted?

21. Unusual dice, usual sums. Two ordinary dice each have face labels $1, 2, 3, 4, 5, 6$. Another pair has face labels

$$(1, 2, 2, 3, 3, 4) \qquad \text{and} \qquad (1, 3, 4, 5, 6, 8).$$

All dice are fair and rolled independently. Use generating polynomials to show that the two pairs have the same distribution of sums. Faces with repeated labels are still distinct outcomes.

22. No fixed points. Let D_n count permutations of $\{1, \dots, n\}$ with no fixed points, with $D_0 = 1$. Find the EGF and deduce $D_n = n! \sum_{j=0}^{n} (-1)^j / j!$. *Hint:* exclude cycles of length one.

23. A permutation that undoes itself. An involution is a permutation σ satisfying $\sigma^2 = \mathrm{id}$. Find the EGF for involutions on n labelled elements, and obtain a finite-sum formula for their number. Interpret the summation index combinatorially.

24. A leader in every group. Partition n distinct people into nonempty, unlabelled groups and choose one leader in each group. Let a_n count the resulting objects, with $a_0 = 1$. Find its EGF, and show that, for $n \ge 1$,

$$a_n = \sum_{k=1}^{n} \binom{n}{k} k^{\,n-k}.$$

Why does this also count functions $f : \{1, \dots, n\} \to \{1, \dots, n\}$ satisfying $f \circ f = f$?

25. Odd cycles versus even cycles. For every integer $m \ge 0$, show that, on a set of $2m$ labelled elements, the number of permutations whose cycles all have odd length equals the number whose cycles all have even length. Find their common number. *Hint:* find the two EGFs and compare their even-degree coefficients.